

PART C

THE SUPER TIMELINE & INVESTIGATION CONCLUSION

CYBERSTER BLUE TEAM INTERNSHIP — WEEK 11 — DAYS 79–80
Timeline Window: 2008-07-20 01:25:00 – 01:30:00 GMT | Filter: m57biz

Case	M57.biz Confidential Data Leak
Methodology	Start Big → Filter Down (date → time window → keyword)
Date Filter	July 20, 2008
Time Window	01:25:00 GMT – 01:30:00 GMT (5-minute window)
Keyword Filter	m57biz
Events in Window	11 filtered timeline entries
Tools Used	Autopsy 4.23.0 Timeline Analysis
Final Verdict	Jean is a phishing victim — NOT a malicious insider

1. Super Timeline Methodology

A super timeline consolidates all timestamped artefacts from the forensic image — file system MAC times, registry last-write times, event log entries, email records, prefetch timestamps, and LNK creation times — into a single chronologically ordered dataset. The full timeline for a Windows system can contain millions of entries. A structured filtering methodology is essential.

Filtering Steps Applied:

- **Step 1 — Date filter:** Narrowed to July 20, 2008 only
- **Step 2 — Time window filter:** 01:25:00 to 01:30:00 GMT (5-minute window around target timestamp)
- **Step 3 — Keyword filter:** 'm57biz' — isolates events directly referencing the target file
- **Result:** 11 timeline entries providing a precise, second-level narrative of the incident

2008-07-20 01:00:00 to 2008-07-20 02:00:00				11 Results
Table	Thumbnail	Summary		
				Save Table as CSV
Icon	Date/Time	Description	Event Type	
	2008-07-20 01:22:45	Sent from: alison@m57.biz <tuckgorge@gmail.com> Sent to: jean@m57.biz : Please send me the information ...	Email Sent	
	2008-07-20 01:22:45	Message from: alison@m57.biz <tuckgorge@gmail.com> To: jean@m57.biz : Please send me the information ...	Email Received	
	2008-07-20 01:26:14	RUNDLL32.EXE : : Prefetch File	Program Run	
	2008-07-20 01:26:18	iCreate Technologies Corp. : Flash Disk 256 MB : 6&3855be95&0&1	Devices Attached	
	2008-07-20 01:27:40	EXCELEXE : : Prefetch File	Program Run	
	2008-07-20 01:27:42	C:\Documents and Settings\Jean\Desktop\m57biz.xls	Recent Documents	
	2008-07-20 01:27:42	C:\Documents and Settings\Jean\Local Settings\Temp	Recent Documents	
	2008-07-20 01:28:00	Sent from: Jean User <jean@m57.biz> Sent to: alison@m57.biz : RE: Please send me the information now : I've ...	Email Sent	
	2008-07-20 01:28:00	Message from: Jean User <jean@m57.biz> To: alison@m57.biz : RE: Please send me the information now : I've ...	Email Received	
	2008-07-20 01:28:04	C:\Documents and Settings\Jean\Desktop	Recent Documents	
	2008-07-20 01:28:04	C:\Documents and Settings\Jean\Desktop\m57biz.xls	Recent Documents	

Figure 1 — Super timeline filtered view in Autopsy

2. Filtered Timeline — Complete Event Set

The following table presents all 11 timeline entries returned after applying the date, time window, and keyword filters. Events are presented in chronological order with the artefact type and forensic interpretation for each.

Timestamp (GMT)	Event Type	Artefact / Description	Interpretation
2008-07-20 01:22:45	Email Received	From: tuckgorge@gmail.com (disp: alison@m57.biz) Subj: Please send me the information	Phishing email received — attacker impersonates Alison
2008-07-20 01:22:45	Email Sent	Same message — dual record in PST	Autopsy records both sent and received copies of the message
2008-07-20 01:26:14	Program Run	RUNDLL32.EXE — Prefetch File	System process execution — may correlate with USB
2008-07-20 01:26:18	Device Attached	iCreate Technologies Corp. Flash Disk 256 MB ID: 6&3855be95&0&1	USB flash drive connected to the system
2008-07-20 01:27:40	Program Run	EXCEL.EXE — Prefetch File	Microsoft Excel launched by Jean
2008-07-20 01:27:42	Recent Document	C:\Documents and Settings\Jean\Desktop\m57biz.xls	Office Recent LNK created at 01:27:42
2008-07-20 01:27:42	Recent Document	C:\Documents and Settings\Jean\Local Settings\Temporary Internet Files\Content.IE5\...	Temporary Internet Files directory activity correlated with file operation
2008-07-20 01:28:00	Email Sent	Jean → alison@m57.biz (actual: tuckgorge@gmail.com) Attachment: m57biz.xls	Spreadsheet transmitted — sent to attacker's Gmail
2008-07-20 01:28:00	Email Record	Duplicate PST record of outbound email	Outlook PST records both outbox and sent copies
2008-07-20 01:28:04	Recent Document	C:\Documents and Settings\Jean\Desktop\Windows updates	Windows updates Desktop folder access record
2008-07-20 01:28:04	Recent Document	C:\Documents and Settings\Jean\Desktop\Windows Updates	Windows Updates Desktop folder LNK created — confirms final

3. Four Core Timeline Events

The following four events constitute the evidential centrepiece of the investigation. Together they establish an unambiguous, second-level sequence from file access to email dispatch.

#	Timestamp (GMT)	Event Type	Artefact Path	Interpretation
1	2008-07-20 01:27:40	Program Run	WINDOWS/Prefetch/EXCEL.EXE-1C75F8D6.pf	Microsoft Excel launched — user initiated file
2	2008-07-20 01:27:42	Recent Document	Application Data/Microsoft/Office/Recent/m57biz.xls	m57biz.xls opened in Excel — Office LNK cre
3	2008-07-20 01:28:00	Email Sent	Outlook PST — jean@m57.biz → tuckgorge@gmail.com	Spreadsheet transmitted to attacker's externa
4	2008-07-20 01:28:04	Recent Document	Documents and Settings/Jean/Recent/m57biz.xls	Windows Recent LNK created — confirms us

4. Second-by-Second Sequence of Events

01:22:45 GMT	Phishing email received tuckgorge@gmail.com (disguised as alison@m57.biz) sends email requesting employee PII including SSNs. Jean sees the display name Alison and believes it is her company President.
01:26:14 GMT	RUNDLL32.EXE executed System process runs — temporally correlated with subsequent USB attachment. May relate to device driver loading or automated system action.
01:26:18 GMT	USB flash drive attached iCreate Technologies Corp. Flash Disk 256 MB (ID: 6&3855be95&0&1) connected. Occurs approximately 86 seconds before Excel is launched. No direct file copy evidence identified, but timing is suspicious.
01:27:40 GMT	Excel.exe launched Microsoft Excel executable launches. Prefetch file EXCEL.EXE-1C75F8D6.pf updated. Jean is about to open the confidential spreadsheet.
01:27:42 GMT	m57biz.xls opened Jean opens m57biz.xls from her Desktop in Excel. Office Recent LNK created. RecentDocs registry key updated. The spreadsheet containing employee names, salaries, and SSNs is now active in Excel.
01:28:00 GMT	Email transmitted Jean sends reply to the phishing email, attaching m57biz.xls. Although the To field displays alison@m57.biz, raw header analysis confirms delivery to tuckgorge@gmail.com. The confidential spreadsheet is now in the attacker's possession.

01:28:04 GMT

Windows Recent LNK created

Windows automatically creates the Recent folder shortcut for m57biz.xls.
NTUSER.DAT RecentDocs further updated. The file access sequence is now completely recorded across three independent artefact categories.

5. Final Investigation Conclusion

Factual Findings

The forensic examination of img_nps-2008-jean.E01 identified a phishing email sent to Jean on July 20, 2008 at 01:22:45 UTC. Although the display name and apparent sender address appeared to belong to Alison at M57.biz, raw header analysis of the RFC 2822 message source confirmed the underlying sender address was tuckgorge@gmail.com — an external Gmail account under the attacker's control. The phishing message requested confidential employee information including names, salaries, and Social Security Numbers.

Within approximately five minutes of receiving the phishing message, Jean opened the spreadsheet file m57biz.xls using Microsoft Excel. This activity was corroborated through three independent artefact categories: Windows prefetch recorded Excel executing at 01:27:40 UTC; Microsoft Office created a recent-file LNK at 01:27:42 UTC; and the Windows Recent folder LNK was created at 01:28:04 UTC. The NTUSER.DAT RecentDocs registry key was updated within the same window. At 01:28:00 UTC, Outlook transmitted an outbound email with m57biz.xls attached, addressed to alison@m57.biz in the display field but delivered to tuckgorge@gmail.com as confirmed by raw header analysis.

A USB flash drive (iCreate Technologies Corp., 256 MB) was connected at 01:26:18 UTC — approximately 84 seconds before Excel was launched. No direct evidence of files being copied to this device was identified within the scope of the current investigation. This finding should be flagged for follow-up USB forensic analysis.

Analytical Conclusion

Jean is a victim of spear-phishing — NOT a malicious insider.

The email header evidence establishes that Jean did not send the confidential spreadsheet to any internal M57.biz address. She sent it to tuckgorge@gmail.com — an external account controlled by the attacker — believing she was replying to her company President, Alison. The spoofed display name in the phishing email created this false belief.

The super timeline corroborates Jean's stated account precisely. The sequence of events — phishing received, Excel launched, spreadsheet opened, email sent — is entirely consistent with a user responding to what she understood to be a legitimate request from her superior. There is no evidence of premeditation, staged data exfiltration, deliberate concealment, or prior unauthorised access to the spreadsheet.

The investigation concludes that a successful social engineering attack was perpetrated against Jean by an unknown external party operating via tuckgorge@gmail.com. The attacker impersonated Alison Smith (President of M57.biz), requested sensitive employee PII under false urgency, and successfully obtained a spreadsheet containing employee names, positions, salaries, and Social Security Numbers. Jean's conduct, while technically the proximate cause of the data disclosure, constitutes unwitting participation rather than intentional exfiltration.

Recommended Next Steps

- Forensic analysis of the USB flash drive (iCreate, 256 MB) — determine if any files were copied
- Legal process to Google for subscriber information on tuckgorge@gmail.com
- Review all other M57.biz email accounts for similar phishing emails targeting other staff
- ShellBag analysis to identify any additional file browsing activity in the window
- Outbound network traffic review for the 01:25–01:30 GMT window

- Examine Alison's (President) account for signs of account compromise or impersonation
- Investigate the 'Volturi Enterprises' organisation name for potential threat actor linkage
- Review Windows event logs for any remote access or privilege escalation events